Lablet 3 Solution

Instructions

Update configurations by performing the following tasks:

Note: Two virtual servers (VMs) are provided as destinations for tests using ping, telnet, and SSH clients from other devices. Do not change configuration on these hosts.

Task 1: Layer 2 Security

- Configure port-security on access interfaces connected to *dog* and *cat* VLANs.
  - The switch should automatically store and remember learned MAC addresses for port-security.
- Ensure only a single MAC address is allowed on each interface.
- Ensure the following violation behavior for networks.
  - "dog" - drop unknown traffic and log a message.
  - "cat" - error disable interface and log a message.

Task 2: Inter-VLAN Routing

- Create VLAN interfaces to provide inter-VLAN routing on **Sw1** for the networks in the data table
- Use the last host address from each network as the IP address on each VLAN interface.

Task 3: Link Aggregation to Virtualized Host

- Configure an LACP EtherChannel 10 from **Sw1** to **vSwitch** made up of all four interfaces connecting the switches.
  - *Note: Interfaces Ethernet 0/0 - 0/3 connect the switches.*
- Ensure the EtherChannel on **Sw1** only forms if **vSwitch** initiates.
- Verify the EtherChannel is operating as a 802.1Q trunk that allows all VLANs.

Task 4: Static Routing

- Configure a default static route on **Dog1**.
- Configure a network static route to the *raccoon* VLAN network on **Cat1**.
- Configure a static host route to **VM-Rabbit** on **Cat2**.
- Ensure all static routes use next-hop IP addresses to the **Sw1** VLAN interface address configured in Task 2.

Task5: Access Control Lists

- Create a named IPv4 ACL "Task5" on **Sw1**.
- Filter inbound traffic from the *cat* network on **Sw1**.
- Rules:
  - BLOCK ICMP traffic from **Cat2** to all other hosts.
  - BLOCK Telnet traffic from the *cat* network to the *raccoon* network.
  - BLOCK HTTPs traffic from any host to the *rabbit* network.
  - ALLOW ICMP traffic from **Cat1** to all other hosts.
  - ALLOW SSH traffic from the *cat* network to **VM-Rabbit**
  - ALLOW SSH traffic from the *cat* network to **VM-Raccoon**
  - Explicitly BLOCK all other traffic.
- Ensure all access control entries (ACEs) are as specific as possible to match the rules.

Task 1 - Layer 2 Security

Sw1

```
enable
config t
!
interface range e1/1-3
 switchport port-security
 switchport port-security mac-address sticky
!
interface e1/1
 switchport port-security violation restrict
!
interface range e1/2-3
 switchport port-security violation shutdown
!
end
```

Verification

Sw1

```
show port-security


Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
                (Count)       (Count)          (Count)
---------------------------------------------------------------------------
      Et1/1              1            1                  0         Restrict
      Et1/2              1            1                  0         Shutdown
      Et1/3              1            1                  0         Shutdown
---------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 4096


show port-security address


               Secure Mac Address Table
-------------------------------------------------------------------------------
Vlan    Mac Address       Type                          Ports   Remaining Age
                                                                   (mins)
----    -----------       ----                          -----   -------------
   5    aabb.cc00.2500    SecureSticky                  Et1/1        -
   6    aabb.cc00.2600    SecureSticky                  Et1/2        -
   6    aabb.cc00.2700    SecureSticky                  Et1/3        -
-------------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 4096
```

Task 2 - Inter-VLAN Routing

Sw1

```
enable
conf t
!
interface vlan 5
 ip address 172.16.5.46 255.255.255.240
 no shut
interface vlan 6
 ip address 172.16.35.254 255.255.252.0
 no shut
interface vlan 7
 ip address 172.16.128.126 255.255.255.128
 no shut
interface vlan 8
 ip address 172.16.201.62 255.255.255.192
 no shut
!
end
```

Verification

```
show ip interface | inc Vlan|Internet address

Vlan5 is up, line protocol is up
  Internet address is 172.16.5.46/28
Vlan6 is up, line protocol is up
  Internet address is 172.16.35.254/22
Vlan7 is up, line protocol is up
  Internet address is 172.16.128.126/25
Vlan8 is up, line protocol is up
  Internet address is 172.16.201.62/26
```

Task 3 - Link Aggregation to Virtualized Host

Sw1

```
enable
config t
!
interface range e0/0-3
 channel-group 10 mode passive
!
end
```

Verification

Sw1

```
show etherchannel 10 port-channel


                Port-channels in the group:
                ---------------------------

Port-channel: Po10    (Primary Aggregator)

------------
```

```
Age of the Port-channel   = 0d:00h:02m:59s
Logical slot/port   = 16/0          Number of ports = 4
HotStandBy port = null
Port state          = Port-channel Ag-Inuse
Protocol            =   LACP
Port security       = Disabled
Fast-switchover     = disabled
Fast-switchover Dampening = disabled

Ports in the Port-channel:

Index   Load   Port        EC state        No of bits
------+------+------+------------------+-----------
  0     00     Et0/0          Passive            0
  0     00     Et0/1          Passive            0
  0     00     Et0/2          Passive            0
  0     00     Et0/3          Passive            0

Time since last port bundled:    0d:00h:02m:31s     Et0/0
```

```
show interface trunk

Port           Mode             Encapsulation  Status        Native vlan
Po10           on               802.1q         trunking      1

Port           Vlans allowed on trunk
Po10           1-4094

Port           Vlans allowed and active in management domain
Po10           1,5-8

Port           Vlans in spanning tree forwarding state and not pruned
Po10           1,5-8
```

Task 4 - Static Routing

> Note: There is "some" connection from this task to Task 2. Namely it uses the same IPs that were determined and used in IPv4 subnetting knowledge. If this was to be a "real" lablet and the goal was to make this task completely independent of Task 2, then complex grading could be done.

Dog1

```
enable
config t
!
ip route 0.0.0.0 0.0.0.0 172.16.5.46
!
end
```

Cat1

```
enable
config t
!
ip route 172.16.201.0 255.255.255.192 172.16.35.254
!
end
```

Cat2

```
enable
config t
!
ip route 172.16.128.31 255.255.255.255 172.16.35.254
!
end
```

Verification

Dog1

```
show ip route 0.0.0.0 0.0.0.0

Routing entry for 0.0.0.0/0, supernet
  Known via "static", distance 1, metric 0, candidate default path
  Routing Descriptor Blocks:
  * 172.16.5.46
      Route metric is 0, traffic share count is 1
```

Cat1

```
show ip route 172.16.201.0 255.255.255.192

Routing entry for 172.16.201.0/26
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
  * 172.16.35.254
      Route metric is 0, traffic share count is 1
```

Cat2

```
show ip route 172.16.128.31 255.255.255.255

Routing entry for 172.16.128.31/32
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
  * 172.16.35.254
      Route metric is 0, traffic share count is 1
```

Task 5 - Access Control Lists

Sw1

```
enable
config t
!
no ip access-list extended Task5
ip access-list extended Task5
 deny icmp host 172.16.33.41 any
 deny tcp 172.16.32.0 0.0.3.255 172.16.201.0 0.0.0.63 eq 23
 deny tcp any 172.16.128.0 0.0.0.127 eq 443
 permit icmp host 172.16.32.41 any
 permit tcp 172.16.32.0 0.0.3.255 host 172.16.128.31 eq 22
 permit tcp 172.16.32.0 0.0.3.255 host 172.16.201.31 eq 22
```

```
 deny ip any any
 exit
!
interface vlan 6
 ip access-group Task5 in
end
```

Verification

Sw1

```
show access-list Task5

Extended IP access list Task5
    10 deny icmp host 172.16.33.41 any (5 matches)
    20 deny tcp 172.16.32.0 0.0.3.255 172.16.201.0 0.0.0.63 eq telnet (1 match)
    30 deny tcp any 172.16.128.0 0.0.0.127 eq 443
    40 permit icmp host 172.16.32.41 any (10 matches)
    50 permit tcp 172.16.32.0 0.0.3.255 host 172.16.128.31 eq 22 (73 matches)
    60 permit tcp 172.16.32.0 0.0.3.255 host 172.16.201.31 eq 22 (37 matches)
    70 deny ip any any (4 matches)
```

```
show ip int vlan 6  | inc access list

  Outgoing Common access list is not set
  Outgoing access list is not set
  Inbound Common access list is not set
  Inbound  access list is Task5
```

Additional Tests

These will only work if the other Tasks were completed successfully. They can be used for additional checks, but the above verification should be use for independent grading.

Combine commands below with `show ip access-list Task5` on **Sw1** to check counters.

Cat1

```
! Should fail due to ACL
Cat1>telnet 172.16.201.31

Trying 172.16.201.31 ...
% Destination unreachable; gateway or host down

! Should work
Cat1>ssh -l cisco 172.16.201.31

Password: cisco

VM-Raccoon>

! Should Work
Cat1>ping 172.16.201.31
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.201.31, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
```

Cat2

```
! Should fail due to ACL
Cat2>ping 172.16.128.31

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.128.31, timeout is 2 seconds:
U.U.U
Success rate is 0 percent (0/5)

! Should work
Cat2>ssh -l cisco 172.16.128.31
Password: cisco

VM-Rabbit>

! Should fail due to explicit deny
Cat2>telnet 172.16.128.31

Trying 172.16.128.31 ...
% Destination unreachable; gateway or host down

! Testing HTTPs: Should fail due to ACL
Cat2>telnet 172.16.128.31 443

Trying 172.16.128.31, 443 ...
% Destination unreachable; gateway or host down
```